

N. GREGORY

MANKIW

PRINCIPLES OF

ECONOMICS

Eighth Edition



CHAPTER

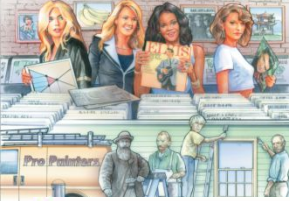
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Application: The Costs of Taxation

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Look for the answers to these questions:

- How does a tax affect consumer surplus, producer surplus, and total surplus?
- What is the deadweight loss of a tax?
- What factors determine the size of this deadweight loss?
- How does tax revenue depend on the size of the tax?



- A tax
 - Drives a wedge between the price buyers pay and the price sellers receive
 - Raises the price buyers pay and lowers the price sellers receive
 - Reduces the quantity bought and sold
- These effects are the same
 - Whether the tax is imposed on buyers or sellers

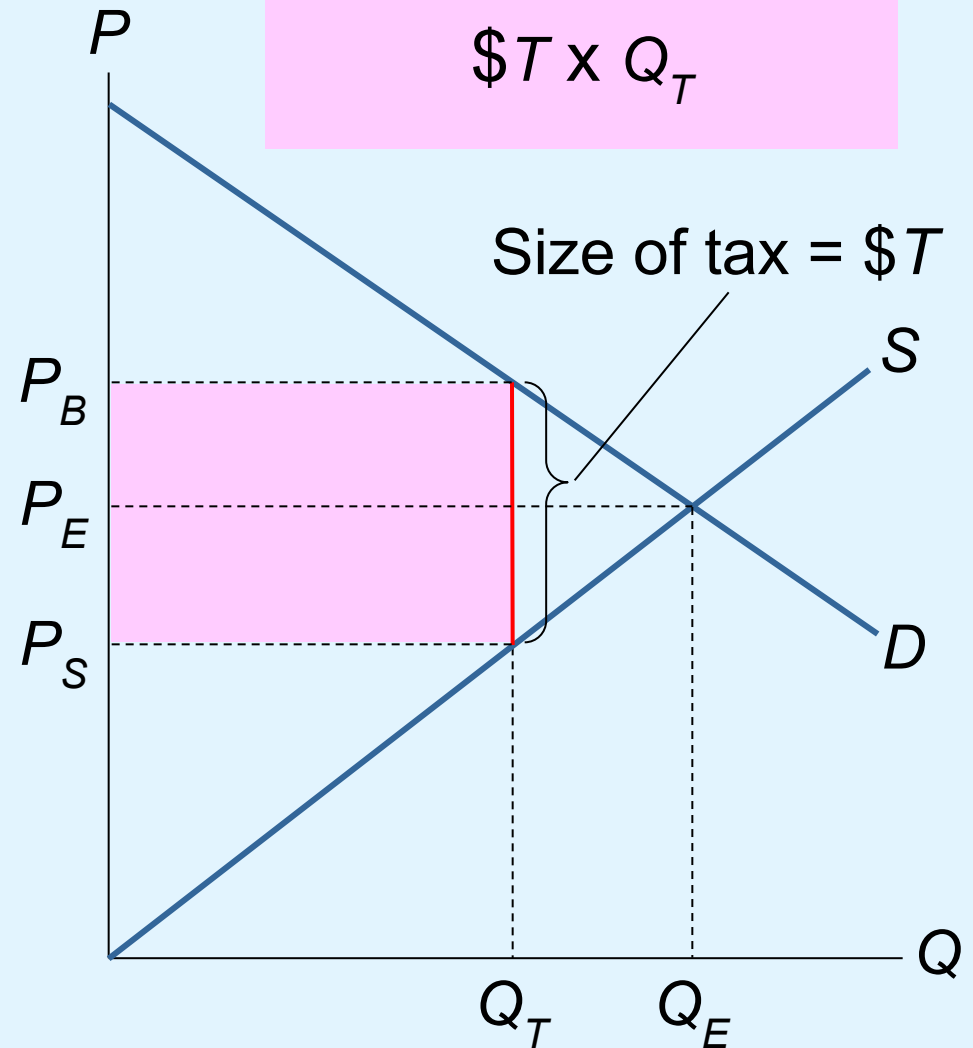
The Effects of a Tax

Equilibrium with no tax:

- Price = P_E
- Quantity = Q_E

Equilibrium with tax = $\$T$ per unit:

- Buyers pay P_B
- Sellers receive P_S
- Quantity = Q_T



The Effects of a Tax

Without a tax,

$$CS = A + B + C$$

$$PS = D + E + F$$

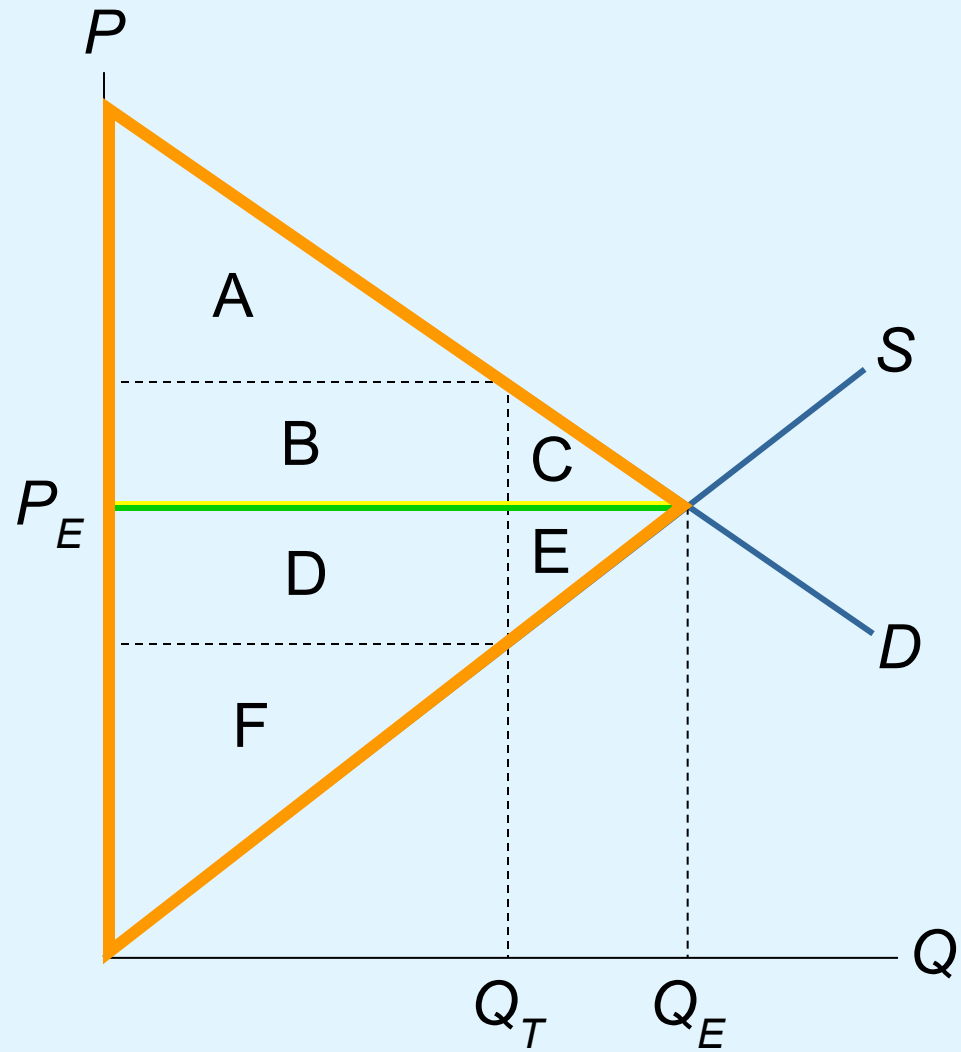
$$\text{Tax revenue} = 0$$

Total surplus

$$= CS + PS$$

$$= A + B + C$$

$$+ D + E + F$$



The Effects of a Tax

With the tax,

$$CS = A$$

$$PS = F$$

$$\text{Tax revenue} = B + D$$

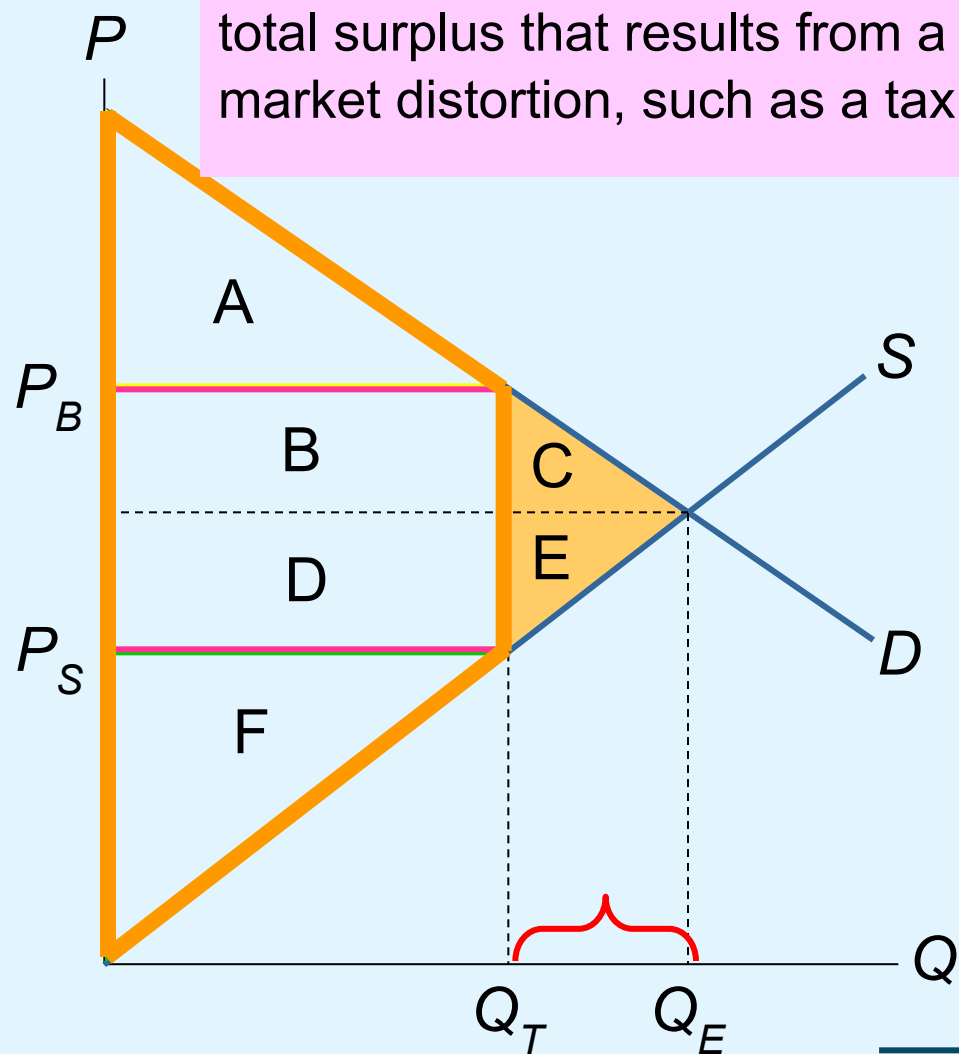
Total surplus

$$= CS + PS$$

$$= A + B + D + F$$

The tax reduces total surplus by $C + E$

$Q_E - Q_T$ = units not sold because of the tax

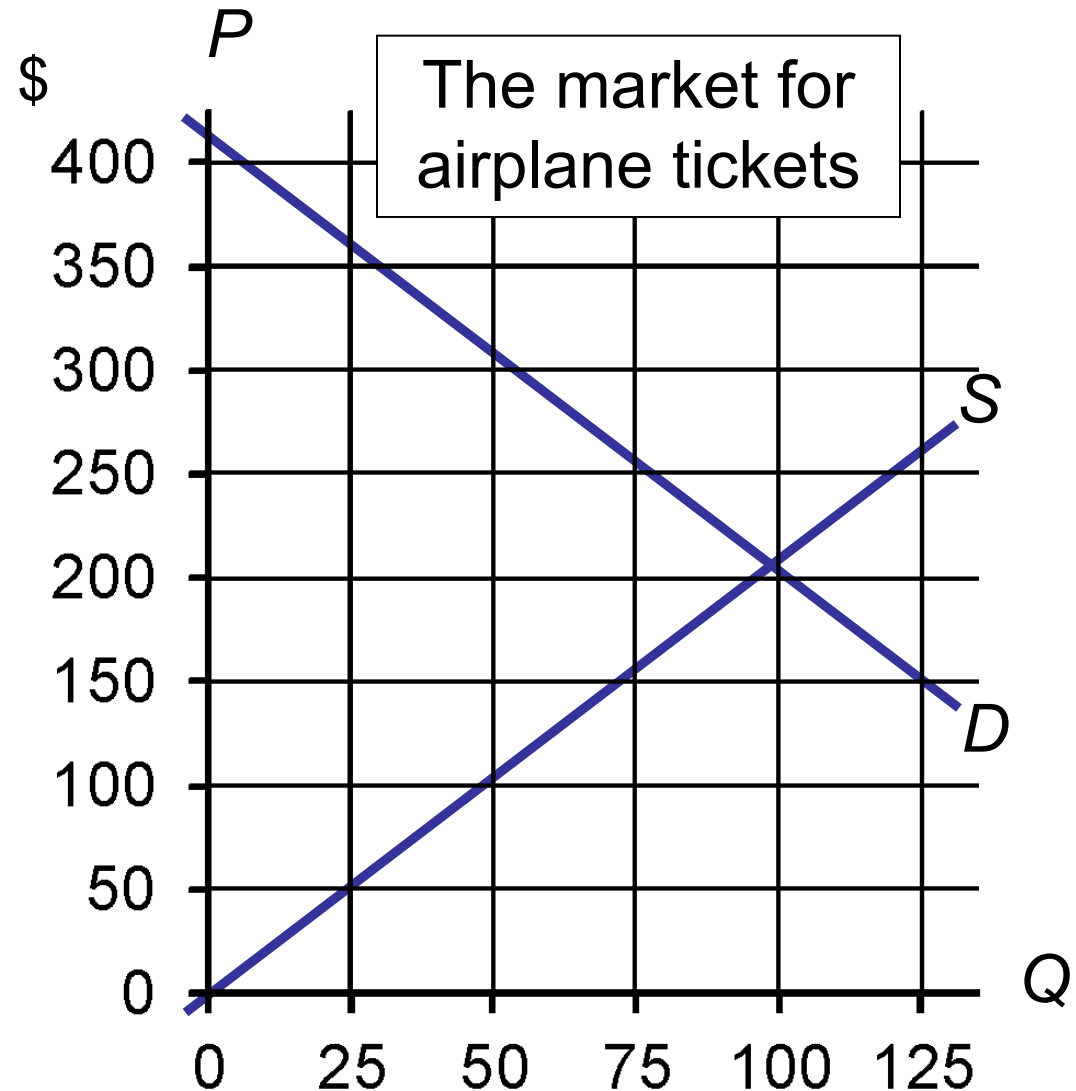


$C + E$ is called the **deadweight loss** (DWL) of the tax, the fall in total surplus that results from a market distortion, such as a tax.

Active Learning 1

Analysis of a tax

- A. Compute CS, PS, and total surplus without a tax.
- B. If \$100 tax per ticket, compute CS, PS, tax revenue, total surplus, and DWL.



Active Learning 1

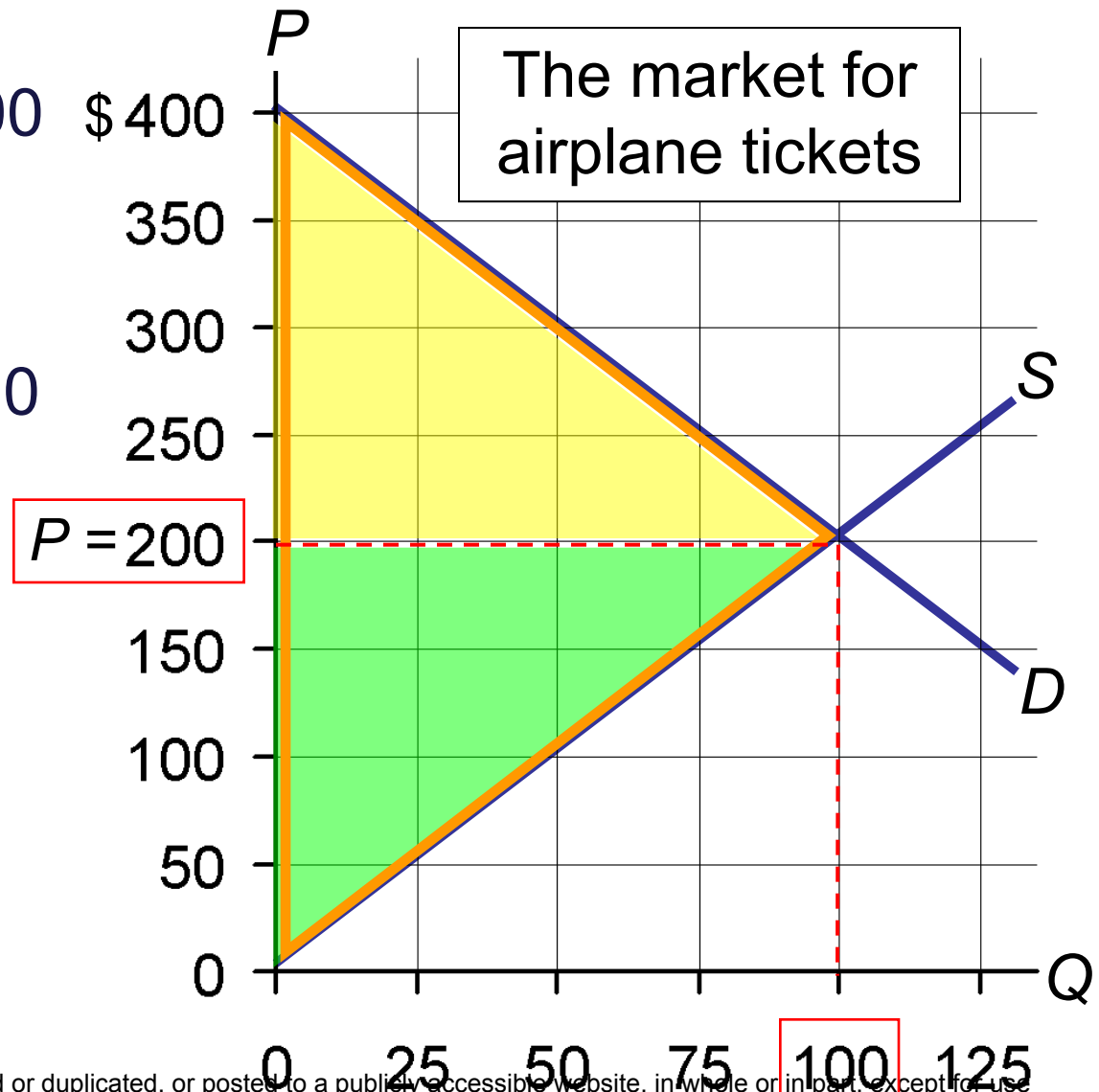
A. Answers

A. Without tax:

$$CS = \frac{1}{2} \times \$200 \times 100 = \$10,000$$

$$PS = \frac{1}{2} \times \$200 \times 100 = \$10,000$$

$$TS = \$20,000$$



Active Learning 1

B. Answers

B. With tax:

$$CS = \frac{1}{2} \times \$150 \times 75$$

$$= \$5,625$$

$$PS = \$5,625$$

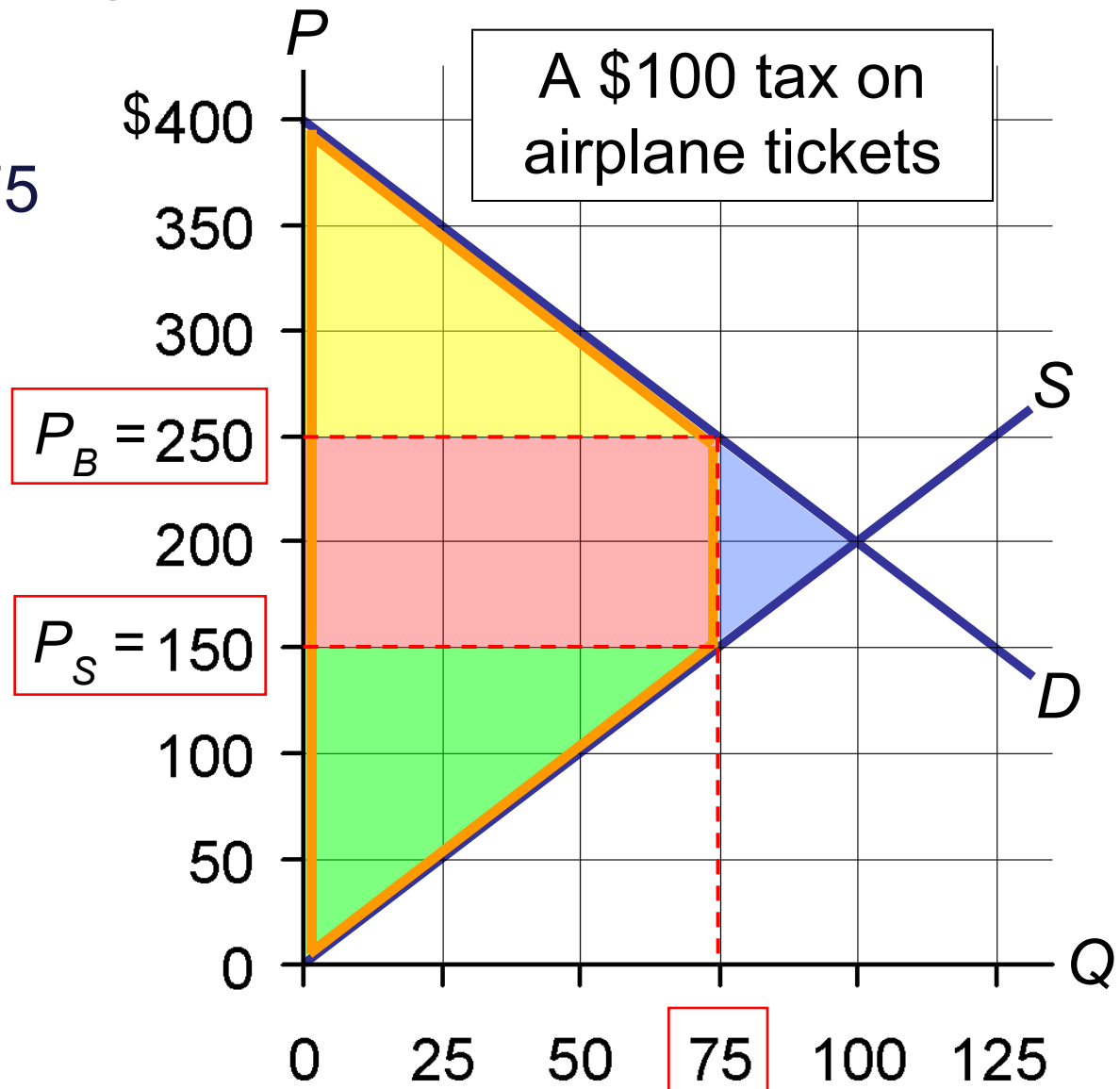
Tax revenue

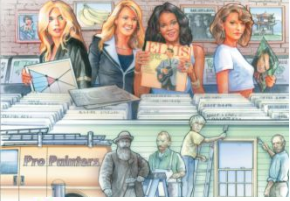
$$= \$100 \times 75$$

$$= \$7,500$$

$$TS = \$18,750$$

$$DWL = \$1,250$$





Determinants of Deadweight Loss

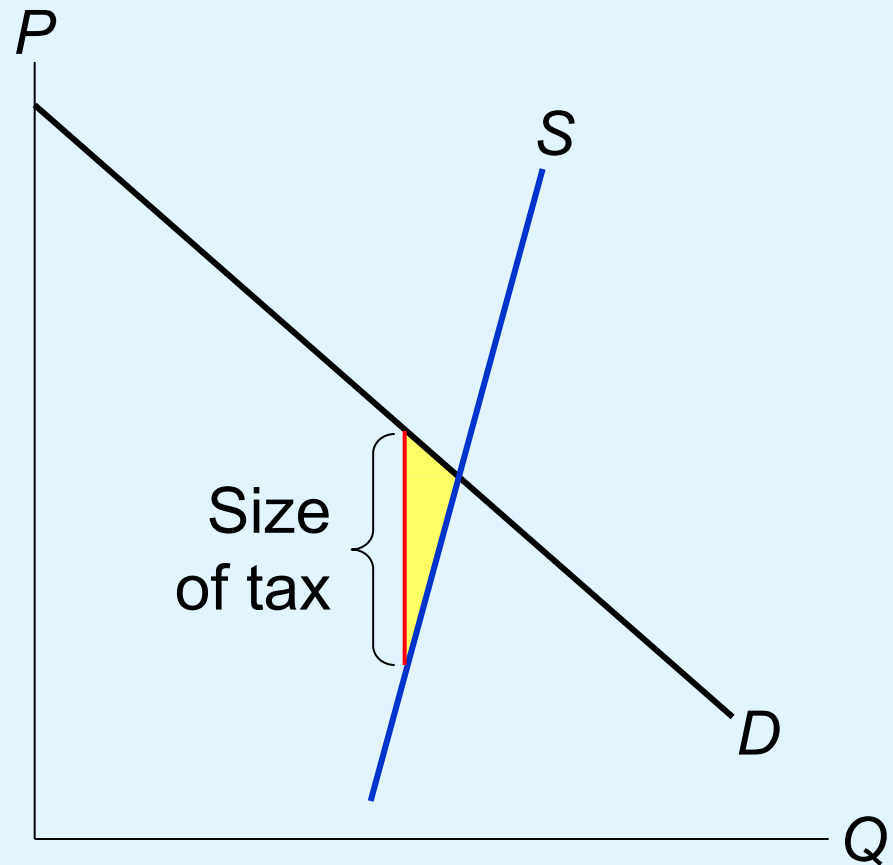
- Price elasticities of supply and demand
 - More elastic supply curve
 - Larger deadweight loss
 - More elastic demand curve
 - Larger deadweight loss
- The greater the elasticities of supply and demand
 - The greater the deadweight loss of a tax

DWL and the Elasticity of Supply

When supply is inelastic,

it's harder for firms to leave the market when the tax reduces P_s .

So, the tax only reduces Q a little, and DWL is small.

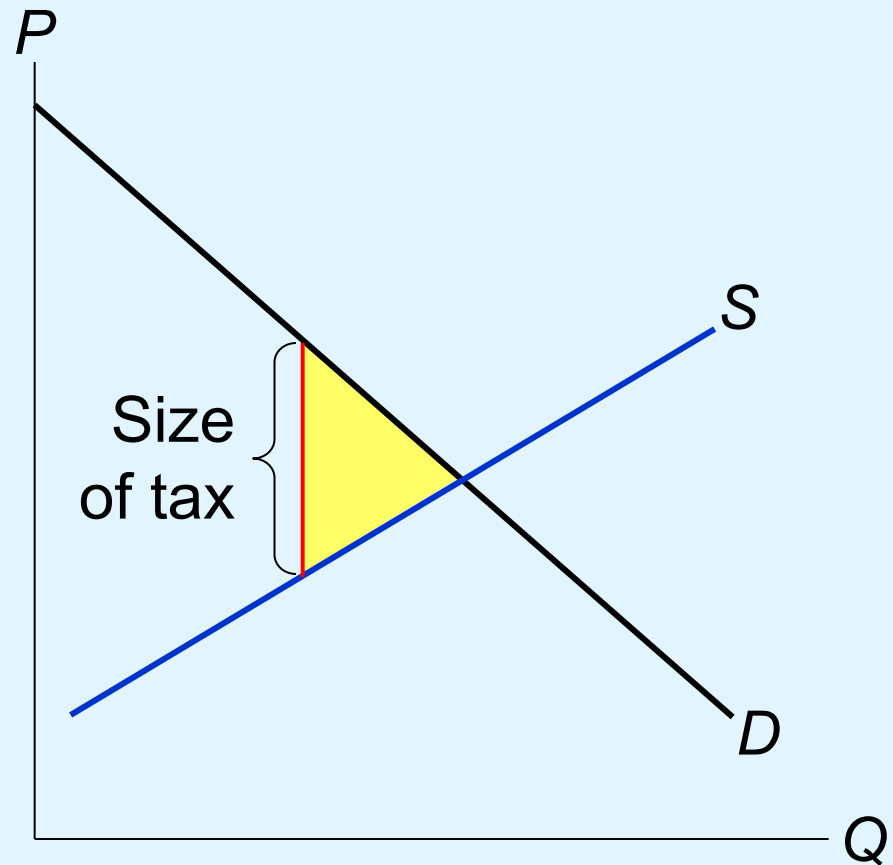


DWL and the Elasticity of Supply

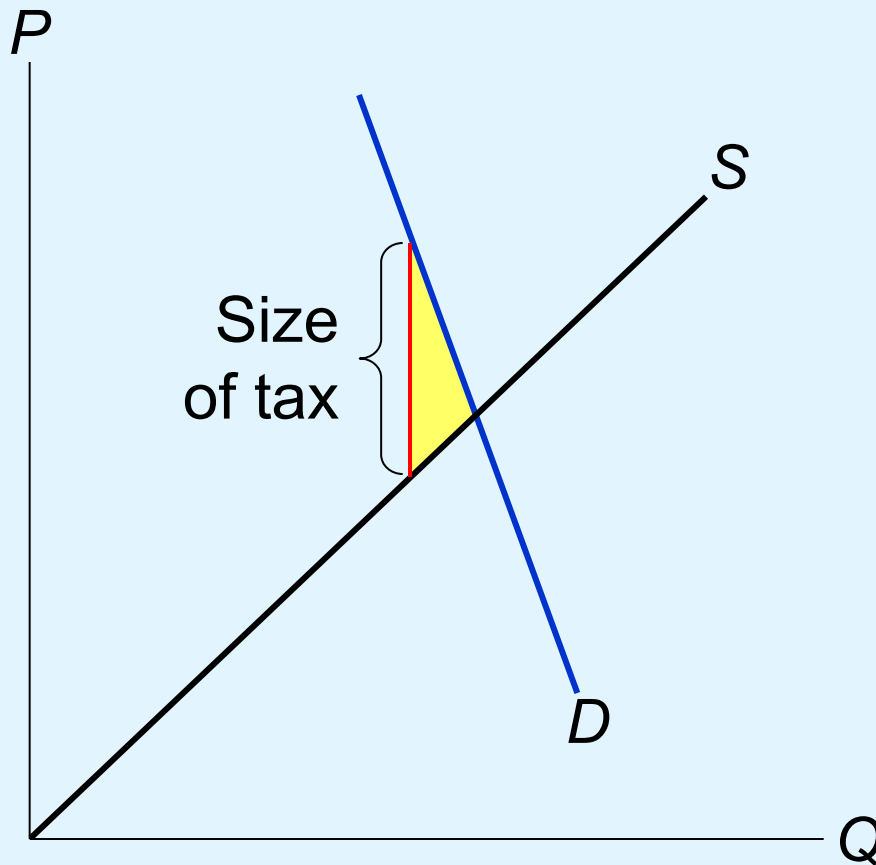
The more elastic is supply,
the easier for firms to leave the market
when the tax reduces P_s ,

the greater Q falls
below the
surplus-maximizing
quantity,

the greater the DWL.



DWL and the Elasticity of Demand

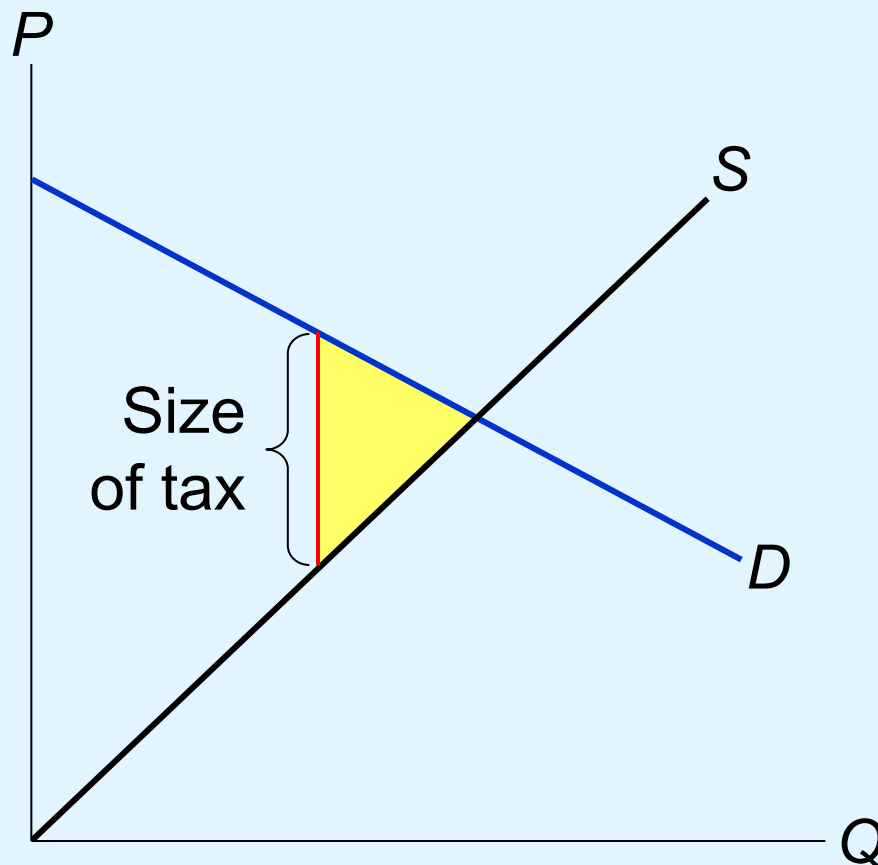


When demand is inelastic,

it's harder for consumers to leave the market when the tax raises P_B .

So, the tax only reduces Q a little, and DWL is small.

DWL and the Elasticity of Demand



The more elastic is demand,
the easier for buyers to
leave the market when
the tax increases P_B ,

the more Q falls below
the surplus-maximizing
quantity,

and the greater the
DWL.

The government must raise tax revenue to pay for schools, police, etc. To do this, it can either tax groceries or meals at fancy restaurants.

- Which should it tax?

How Big Should the Government Be?

- A bigger government
 - Provides more services, but requires higher taxes, which cause DWLs
 - The larger the DWL from taxation, the greater the argument for smaller government
- Tax on labor income - especially important
 - Biggest source of government revenue
- Marginal tax rate: about 40%
- How big is the DWL?

How Big Should the Government Be?

- 40% labor tax - Small or large deadweight loss?

- Some believe labor supply is fairly inelastic

- Almost vertical: most people would work full-time regardless of wage
- Tax on labor: small DWL



“What’s your position on the elasticity of labor supply?”

How Big Should the Government Be?

- Others: labor supply is more elastic
 - Labor taxes are highly distorting: some groups of workers have elastic supply and can respond to incentives
 - Tax on labor: greater DWL
 - Many workers can adjust their hours
 - Some families have 2nd earners; some discretion over whether and how much to work
 - Many of the elderly can choose when to retire
 - Some people work in the “underground economy” to evade high taxes



Effects of Changing the Size of the Tax

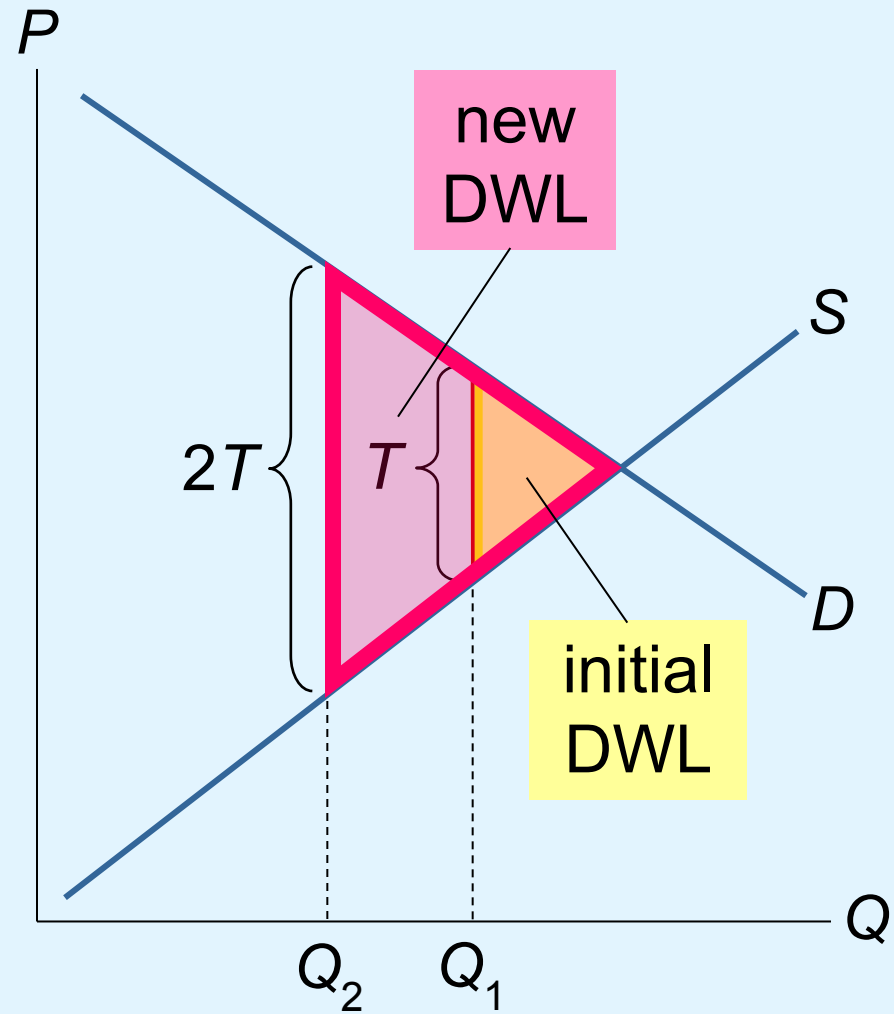
- As the tax increases
 - Deadweight loss increases
 - Even more rapidly than the size of the tax
 - Tax revenue
 - Increases initially
 - Then decreases
 - The higher tax: drastically reduces the size of the market

DWL and the Size of the Tax

Initially, the tax is T per unit.

Doubling the tax

causes the DWL to more than double.

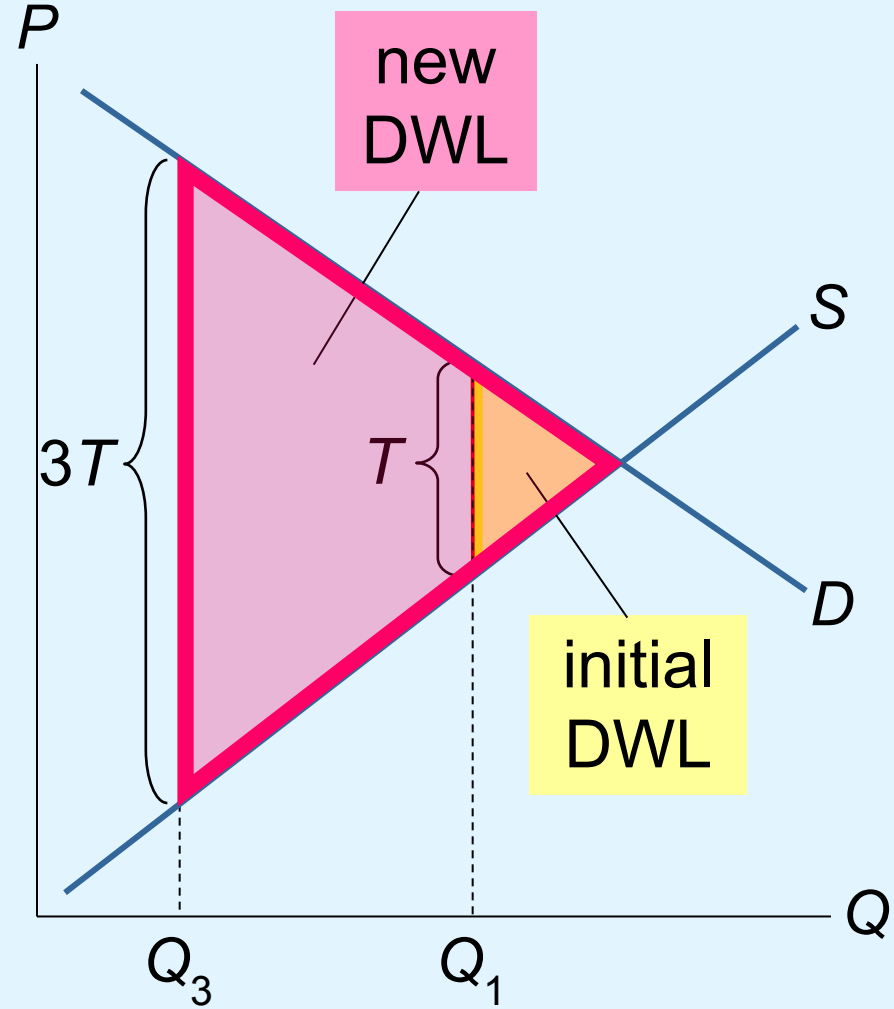


DWL and the Size of the Tax

Initially, the tax is T per unit.

Tripling the tax

causes the DWL to more than triple.



DWL and the Size of the Tax

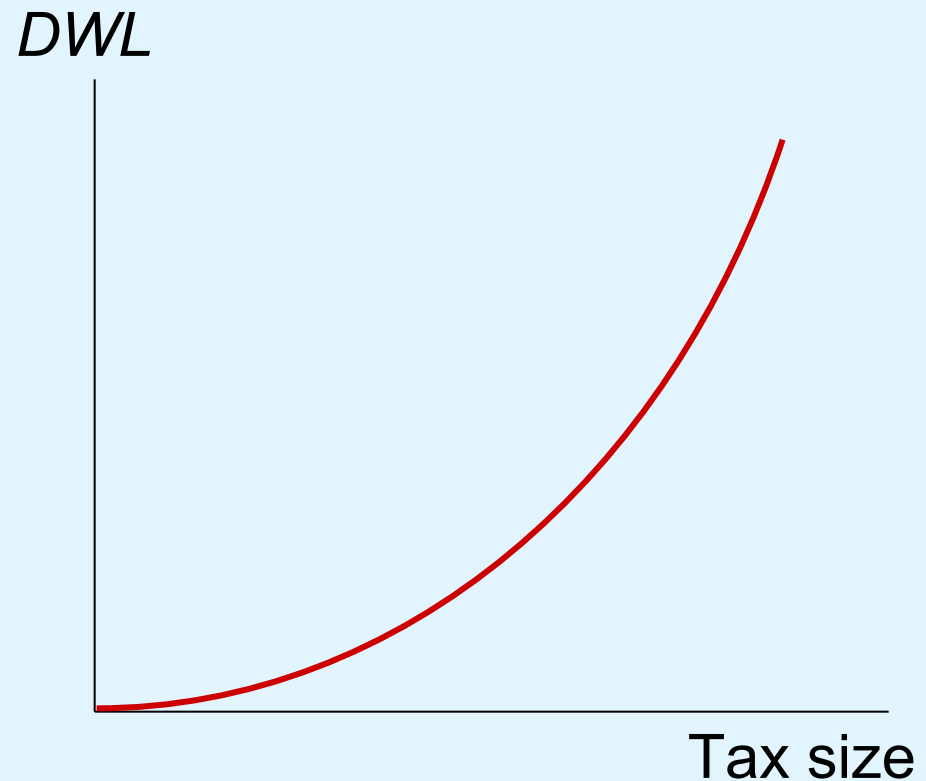
Implication

When tax rates are low, raising them doesn't cause much harm, and lowering them doesn't bring much benefit.

When tax rates are high, raising them is very harmful, and cutting them is very beneficial.

Summary

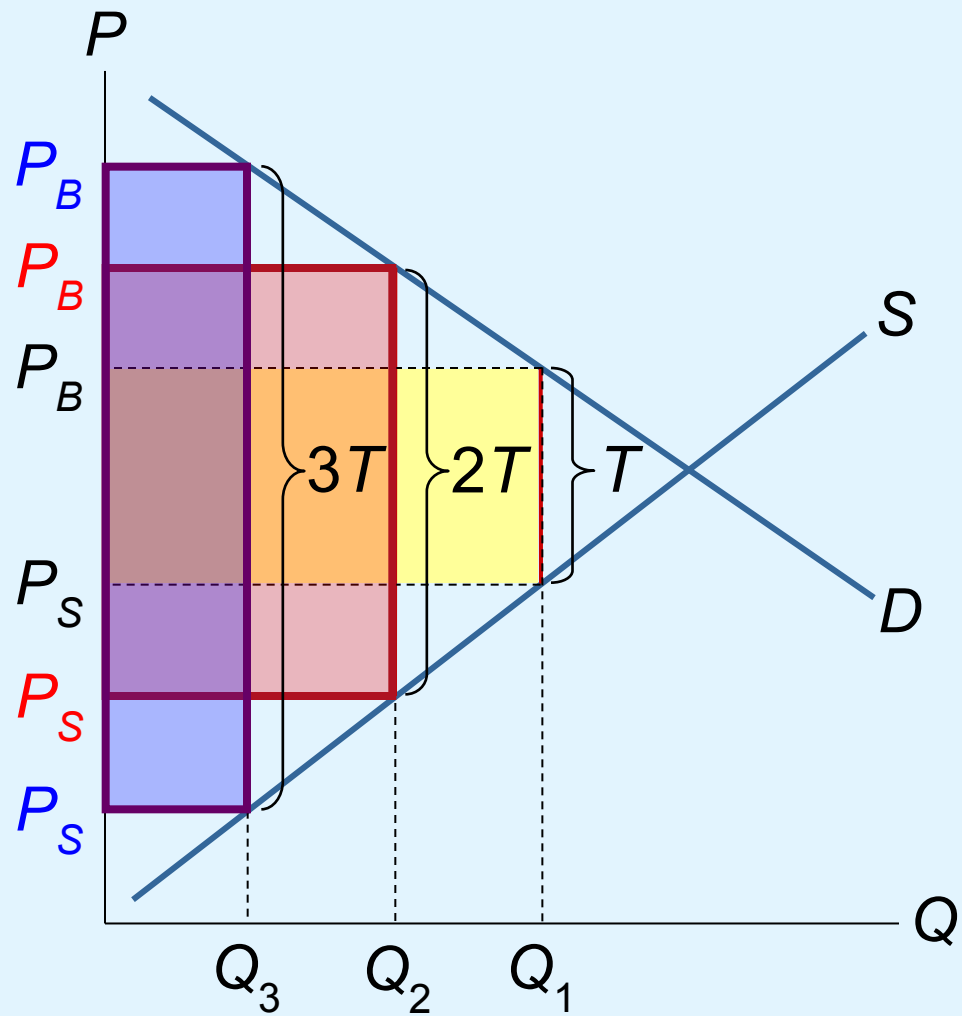
When a tax increases, DWL rises even more.



Revenue and the Size of the Tax

When the tax is small, increasing it causes tax revenue to rise.

When the tax is larger, increasing it causes tax revenue to fall.

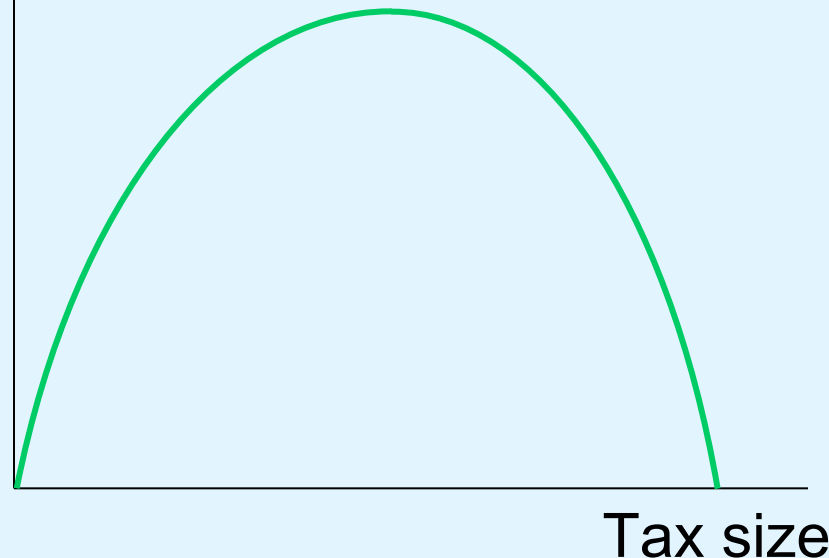


Revenue and the Size of the Tax

The ***Laffer curve*** shows the relationship between the size of the tax and tax revenue.

Tax
revenue

The Laffer curve



Summary

- A tax on a good reduces the welfare of buyers and sellers. This welfare loss usually exceeds the revenue the tax raises for the govt.
- The fall in total surplus (consumer surplus, producer surplus, and tax revenue) is called the deadweight loss (DWL) of the tax.
- A tax has a DWL because it causes consumers to buy less and producers to sell less, thus shrinking the market below the level that maximizes total surplus.

Summary

- The price elasticities of demand and supply measure how much buyers and sellers respond to price changes. Therefore, higher elasticities imply higher DWLs.
- An increase in the size of a tax causes the DWL to rise even more.
- An increase in the size of a tax causes revenue to rise at first, but eventually revenue falls because the tax reduces the size of the market.